

Build a professional-grade "MLB Hitter Lab" application using React, Tailwind CSS, and the Gemini 1.5 Pro API. The app is designed for MLB-level players to analyze high-speed video of their swings.

1. VIDEO ANALYTICS CORE:

- Implement a video upload system that uses Gemini 1.5 Pro's 1-million-token context window to perform "Temporal Reasoning."
- The AI must identify and timestamp four critical phases: Stance/Load, Toe-Touch, Launch/Contact, and Follow-Through.
- Extract Biomechanical Data: Calculate Lead Arm Barring (angle), Hip-Shoulder Separation (X-Factor), and Front Knee Bracing.

1. STATCAST METRIC ESTIMATION:

- Add a "Calibration Menu" where the user inputs their Bat Length (e.g., 34") and their Height. Use these as reference objects for pixel-to-inch scaling.
- Calculate professional metrics:
 - Attack Angle (vertical angle of the barrel at contact).
 - Vertical Bat Angle (tilt at contact).
 - Estimated Exit Velocity: Use the formula $[Distance\ in\ pixels / (Frames * Frame\ Duration)]$ calibrated against the known bat length.
 - Estimated Distance: Apply a standard physics model (approx. 5ft of distance per 1mph EV at a 25-degree launch angle).

1. MLB COACHING ENGINE:

Make changes, add new features, ask for anything



Generating app
Enjoy these tips while you wait

And please keep track of user data with Firebase Auth to securely identify users.

Use Google Sign-In with Firebase Auth to securely identify users.

Publish to GitHub

When your prototype is ready for a full development environment, publish the entire codebase to a new GitHub repository with a single click.